Java: Exception Handling

"Concept && Coding" YT Video Notes

What is Exception?

- It's an event, that occurs during the execution of the program.
- It's an object, that represents an error.
- It Creates the Exception Object, which contain information about the Error like
  1. Its Type of Exception and Message
  2. Stack trace etc.
- Runtime system see this Exception Object and find the class which can handle it.

With Exception handling:

```java
public void myMethod(int schoolClassNumber) {

    try {
        int noOfStudents = getStudentCapacityofClass(schoolClassNumber);
        String[] names = new String[noOfStudents];
        names[0] = "new value";
    }
    catch (IndexOutOfBoundsException expObj){
        //do something
    }
    catch (Exception expObj){
        //do something
    }
}
```

But remember, exception handling is little expensive, if stack trance is huge and it is not handled or handled at parent class.

```java
public int myMethod(int a, int b) {
    int val;
    try{
         val = a/b;
    } catch (ArithmeticException exp){
        val = -1;
    }
    return val;
}

public int myMethod2(int a, int b) {

        if(b == 0){
            return -1;
        }

        int val = a/b;
        return val;
}
```

Try to avoid using exception handling, if you can. Like in this example, its not required